SC2006 - Software Engineering

Lab 4 Deliverables

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1. Control Class

The control class we used for testing is **AuthController**, which is responsible for managing all user authentication processes within ResaleSense (including both user login and user registration).

When a user login to the application, AuthController receives input credentials consisting of the email and password. It validates that the inputs are correctly formatted and non-empty before proceeding. The controller will retrieve the corresponding user record from the database and compare the provided password with the stored hash password. If the passwords match, authentication is successful. The user will then be directed to the home screen according to their role, either Admin or Consumer. If the email or password is invalid, the system will return an error message. However, if the account is permanently banned, the system will return an error message. In cases of temporary bans, the AuthController will check if the ban period has expired, before allowing them to login.

Apart from user login, AuthController also handles user registration. When a new user registers for the application, the controller ensures that required information is collected, including the name, email, password and confirm password. Each field will be validated for correctness and completeness. The controller checks that the email follows a valid format and is not already associated with an existing account. The password is then securely hashed before being stored in the database. If registration is successful, a new record is created and the system returns a confirmation response indicating successful account creation. However, if any required information is missing or the email is already used, the controller will return an error message.

2. Black Box Testing

I. Equivalence Class Testing

Equivalence class testing is a black-box design technique used to reduce the total number of test cases while ensuring that all possible input conditions are sufficiently covered. In this method, the input data is organized into groups known as equivalence classes, whereby each class represents a category of inputs expected to produce the same system behaviour. One representative value is selected from each class for testing, because if the system behaves correctly for one input in that class, it is assumed to behave similarly for all other inputs within that class.

For AuthController, the login and registration functions have their equivalent classes.

- **Login:** Valid classes include valid email addresses that are registered and their corresponding passwords, whereas invalid classes include invalid emails, emails not registered, incorrect passwords, and accounts that are temporarily banned or permanently banned, causing the system to reject login.
- **Registration:** Valid classes include a unique username, valid email addresses that are not in use, and matching passwords, whereas invalid classes cover missing or duplicate data, invalid email addresses, or passwords that do not meet the requirements.

II. Boundary Value Testing

Boundary Value Testing is a black-box technique that focuses on testing the edges or limits of acceptable input ranges. As errors are most likely to occur at the boundary conditions rather than in the middle of the input range, this method ensures that the system behaves correctly when values are on-the-boundary, just-below or just-above boundaries.

For AuthController, boundary value testing is applied to the password and email fields.

- **Password:** Verifies that the password meets the minimum length requirement are accepted (Valid), whereas those shorter are rejected (Invalid)
- **Email:** Verifies that the email inputs with missing components are correctly identified as invalid.

III. Test Cases and Results

III.I. Login

Inputs: Email and Password

No.	Test Input	Description	Expected Output	Actual Output
1	**(Valid)** Email: “consumer@example.com” **(Valid)** Password: “example123”	Account exists, email and password are correct	Login successful, user directed to homepage	Login successful, user directed to homepage
2	**(Invalid)** Email: “” **(Valid)** Password: “example123”	Empty email field	Login failed, system displays “Please fill in this field.” in the email field	Login failed, system displays “Please fill in this field.” in the email field
3	**(Valid)** Email: “consumer@example.com” **(Invalid)** Password: “”	Empty password field	Login failed, system displays “Please fill in this field.” in the password field	Login failed, system displays “Please fill in this field.” in the password field
4	**(Valid)** Email: “consumer@example.com” **(Invalid)** Password: “wrongpassword”	Account exists, incorrect password entered	Login failed, system displays error message “Invalid email or password”	Login failed, system displays error message “Invalid email or password”
5	**(Invalid)** Email: “consumerexample.com” **(Valid)** Password: “example123”	Invalid email, missing “@” symbol in email field	Login failed, system displays “Please include an ‘@’ in the email address.” ‘consumerexample.com’ is missing an @” in the email field	Login failed, system displays “Please include an ‘@’ in the email address.” ‘consumerexample.com’ is missing an @” in the email field
6	**(Invalid)** Email: “fakeconsumer@example.com” **(Valid)** Password: “fakepassword”	Account does not exist	Login failed, system displays “Invalid email or password.”	Login failed, system displays “Invalid email or password.”
7	**(Invalid)** Email: “banned1@example.com” **(Valid)** Password: “example123”	Account exists, user is temporarily or permanently banned	Login failed, system displays error message “ACCOUNT_BANNED”	Login failed, system displays error message “ACCOUNT_BANNED”
8	**(Valid)** Email: “banned1@example.com”	Account exists, banned has been lifted	Login successful, user directed to the home page	Login successful, user directed to the home page

	(Valid) Password: “example123”			
9	**(Valid)** Email: “consumer@example.com” **(Invalid)** Password: “exa”	Password is shorter than 6 characters	Login failed, system displays error message “Invalid email or password.”	Login failed, system displays error message “Invalid email or password.”

Results of Testing (Login): All login test cases passed successfully, confirming that the system correctly authenticates valid users and rejects invalid or banned accounts.

III.II. Sign Up

Inputs: Username, Email, Password and Confirm Password

No.	Test Input	Description	Expected Output	Actual Output
1	**(Valid)** Username: “consumer3” **(Valid)** Email: “consumer3@example.com” **(Valid)** Password: “example123” **(Valid)** Confirm Password: “example123”	All fields are valid, username is unique, email is not in use, password and confirm password matches	Registration successful, confirmation message “User registered successfully!” displayed	Registration successful, confirmation message “User registered successfully!” displayed
2	**(Invalid)** Username: “” **(Valid)** Email: “consumer4@example.com” **(Valid)** Password: “example123” **(Valid)** Confirm Password: “example123”	Missing name field	Registration failed, system displays “Please fill in this field.” in the username field	Registration failed, system displays “Please fill in this field.” in the username field
3	**(Valid)** Username: “consumer4” **(Invalid)** Email: “” **(Valid)** Password: “example123” **(Valid)** Confirm Password: “example123”	Missing email field	Registration failed, system displays “Please fill in this field.” in the email field	Registration failed, system displays “Please fill in this field.” in the email field
4	**(Valid)** Username: “consumer4” **(Valid)** Email:	Missing password field	Registration failed, system displays “Please fill in this	Registration failed, system displays “Please fill in this

	“consumer4@example.com” **(Invalid)** Password: “” **(Valid)** Confirm Password: “example123”		field.” in the password field	field.” in the password field
5	**(Valid)** Username: “consumer4” **(Valid)** Email: “consumer4@example.com” **(Valid)** Password: “example123” **(Invalid)** Confirm Password: “”	Missing confirm password field	Registration failed, system displays “Please fill in this field.” in the confirm password field	Registration failed, system displays “Please fill in this field.” in the confirm password field
6	**(Invalid)** Username: “consumer2” **(Valid)** Email: “consumer4@example.com” **(Valid)** Password: “example123” **(Valid)** Confirm Password: “example123”	Username is already taken	Registration failed, system displays error message “Registration failed. This username is already taken.”	Registration failed, system displays error message “Registration failed. This username is already taken.”
7	**(Valid)** Username: “consumer4” **(Invalid)** Email: “consumer2@example.com” **(Valid)** Password: “example123” **(Valid)** Confirm Password: “example123”	Email is already in use	Registration failed, system displays error message “Registration failed. This email is already in use.”	Registration failed, system displays error message “Registration failed. This email is already in use.”
8	**(Valid)** Username: “consumer4” **(Valid)** Email: “consumer4@example.com” **(Invalid)** Password: “exa” **(Valid)** Confirm Password: “exa”	Password does not meet requirements	Registration failed, system displays error message “Password must be at least 6 characters long.”	Registration failed, system displays error message “Password must be at least 6 characters long.”
9	**(Valid)** Username: “consumer4” **(Valid)** Email: “consumer4@example.com”	Password and confirm password not matching	Registration failed, system displays error message “Passwords do not match”	Registration failed, system displays error message “Passwords do not match”

	(Valid) Password: “example123” **(Invalid)** Confirm Password: “example1234”			
10	**(Valid)** Username: “consumer4” **(Invalid)** Email: “consumer4example.com” **(Valid)** Password: “example123” **(Valid)** Confirm Password: “example123”	Invalid email, missing “@” symbol in email field	Registration failed, system displays “Please include an ‘@’ in the email address. ‘consumer4example.com’ is missing an ‘@’.” in the email field	Registration failed, system displays “Please include an ‘@’ in the email address. ‘consumer4example.com’ is missing an ‘@’.” in the email field

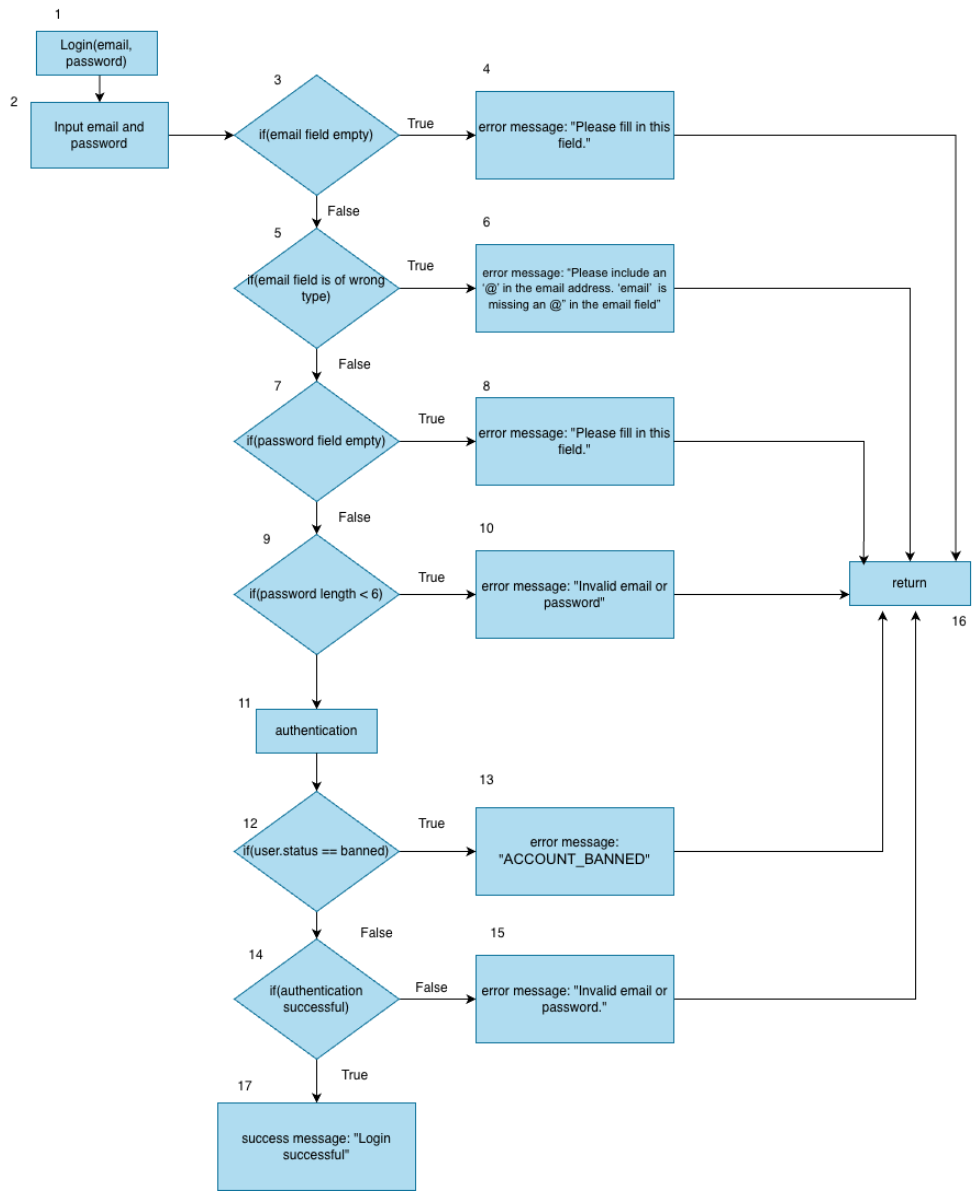
Results of Testing (Sign Up): All registration test cases have passed successfully. The system returns a success message for valid input data, whereas invalid or duplicate entries return error messages according to the requirements.

3. White Box Testing

I. Login

I.I. Control Flow Graph

The login() function in the AuthController handles user authentication. It verifies if the email and password fields are filled, checks that the email is in a valid format, ensures the password meets minimum length requirements, and authenticates the credentials against the database. Depending on these checks, it either grants access or displays an appropriate error message.



I.II. Basic Path Testing

Cyclomatic Complexity = | decision points | + 1 = 6+1 = 7

Basis Paths:

1. Baseline path (Successful): 1, 2, 3, 5, 7, 9, 11, 12, 13, 17
2. Basis path 2 (Empty email): 1, 2, 3, 4, 16
3. Basis path 3 (Invalid email format): 1, 2, 3, 5, 6,16
4. Basis path 4 (Empty password): 1, 2, 3, 5, 7, 8, 16
5. Basis path 5 (Password too short): 1, 2, 3, 5, 7,9 ,10, 16
6. Basis path 6 (Authentification failed): 1, 2, 3, 5, 7, 9, 11, 12, 14, 15, 16
7. Basis path 7 (Banned account): 1, 2, 3, 5, 7, 9, 11, 12, 13, 16

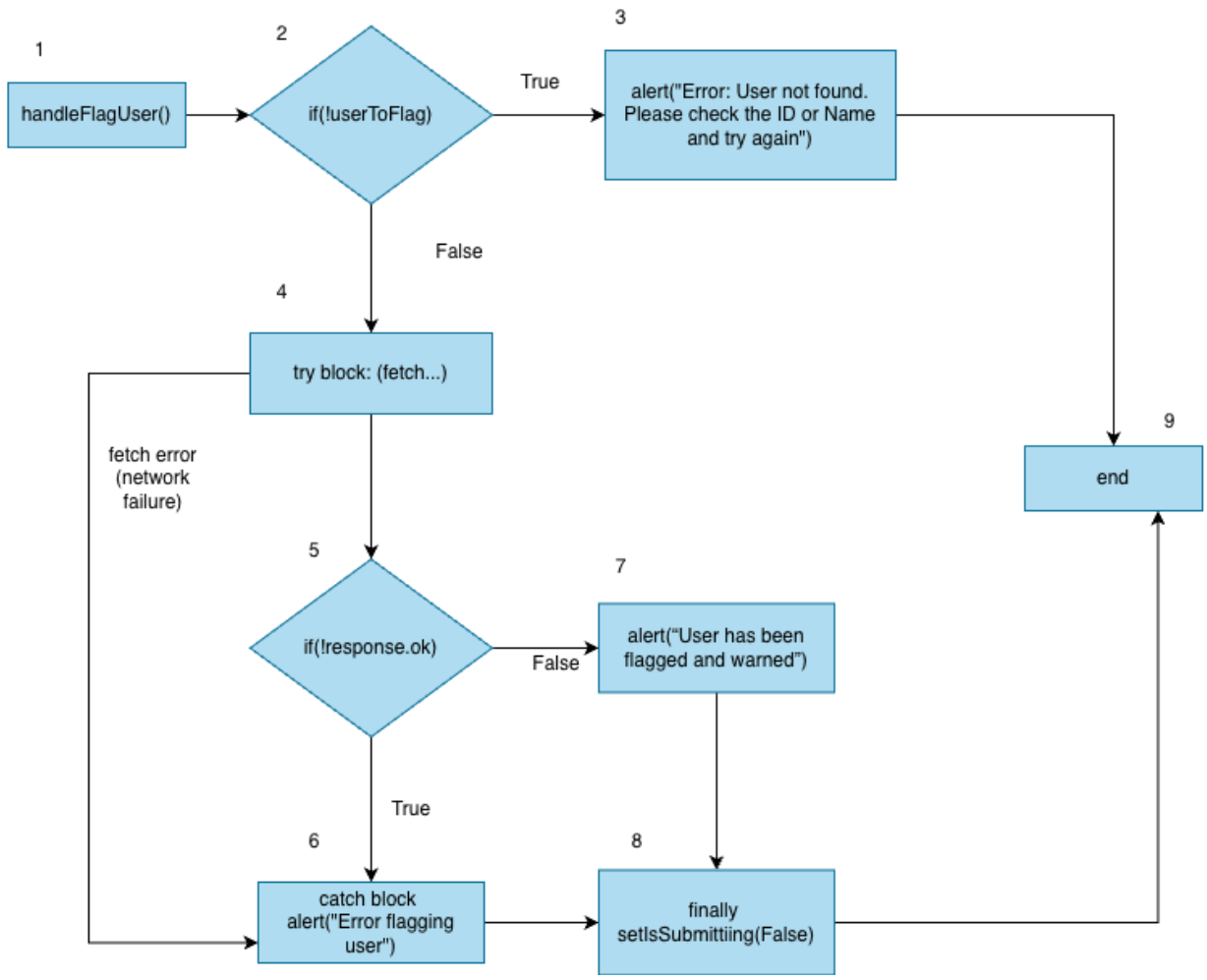
I.III. Test Cases and Results

No.	Test Input	Expected Output	Actual Output	Pass?
1	Email: “admin@example.com” Password: “123123”	Login successful, user directed to homepage	Login successful, user directed to homepage	Yes
2	Email: “” Password: “123123”	“Please fill in this field.”	“Please fill in this field.”	Yes
3	Email: “admin@example.com” Password: “”	“Please fill in this field.”	“Please fill in this field.”	Yes
4	Email: “admin@example.com” Password: “123”	“Invalid email or password”	“Invalid email or password”	Yes
5	Email: “adminexample.com” Password: “123123”	“Please include an ‘@’ in the email address. ‘adminexample.com’ is missing an @” in the email field”	“Please include an ‘@’ in the email address. ‘adminexample.com’ is missing an @” in the email field”	Yes
6	Email: “fakeadmin.com” Password: “fakeadmin”	“Invalid email or password”	“Invalid email or password”	Yes
7	Email: “bannedacc.com” Password: “bannedacc”	“ACCOUNT_BANNED”	“ACCOUNT_BANNED”	Yes

II. handleFlagUser

II.I. Control Flow Graph

The handleFlagUser() function is responsible for processing an admin's action to flag or ban a user. It performs several conditional checks, verifying whether the user exists, determining if they are already flagged, asking for admin confirmation, and validating the backend response. Because it includes multiple nested decisions and error-handling structures, this function provides rich branch coverage for white-box testing.



II.II. Basic Path Testing

Cyclomatic Complexity = | decision points | + 1 = 3+1 = 4

Basis Paths:

1. Baseline path (Successful): 1, 2, 4, 5, 7, 8, 9
2. Basis path 2 (User not found): 1, 2, 3, 9
3. Basis path 3 (Network error): 1, 2, 4, 6, 8, 9
4. Basis path 4 (API error): 1, 2, 4, 5, 6, 8, 9

II.III. Test Cases and Results

No.	Test Input	Expected Output	Actual Output	Pass?
1	UserID or name: fakeid Reason for Flagging: fakereason	“Error: User not found. Please check the ID or Name and try again”	“Error: User not found. Please check the ID or Name and try again”	Yes
2	UserID or name: exampleUser Reason for Flagging: Spam (User is not flagged yet)	“User has been flagged and warned”	“User has been flagged and warned”	Yes
3	UserID or name: exampleUser Reason for Flagging: Spam (User is not flagged yet) Disconnect WIFI	“Something went wrong”	“Something went wrong”	Yes
4	UserID or name: example2User Reason for Flagging: Spam Error in API by stopping backend	“Error flagging User”	“Error flagging User”	Yes